Optimization - 4 ⇒

Agenda ⇒
- → Recap
- → Lagrange multiplier
- → Unconstraint optimization
- → constraint optimization

(*) (*)

Detailed Recap: $f(x)$ → Derivative

(a) Derivative: $\frac{d}{dx} f(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$.

(b) Gradient descent update rule: ⇒ GD

Problem: $\min_x f(x)$ (*) minimization problem

(1) $x^{(0)}$ → Initial guess for x.

(2) $x^{(t+1)} = x^{(t)} - \eta \cdot \frac{\partial f(x)}{\partial x}$ → Partial derivative; → Learning rate

(c) Partial derivative: $f(x, y, z)$.

$\frac{d}{dx} f(x,y,z)$; $\frac{\partial}{\partial x} f(x,y,z)$.

cannot treat y, z as constants. — y, z are treated as constants.

$\left[\frac{\partial f}{\partial x}\right]$ ⇒ y, z would remain constant

$\frac{\partial f}{\partial y}$ ⇒ x, z

(*) Multivariate functions

(*) (*)

Maxima / Minima ⇒

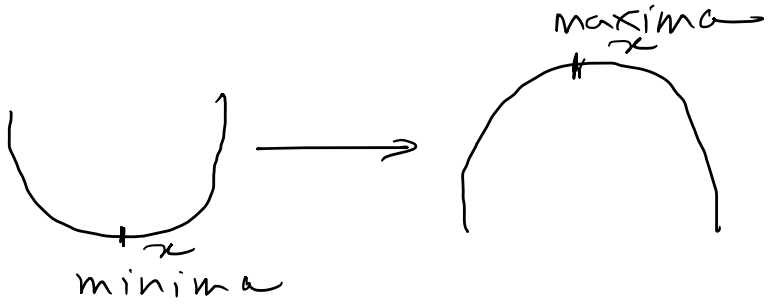
(*) To find candidate / critical point

* $f(x)$ → $f'(x) = 0$ ⇒ [maxima or minima]

if $f''(x) > 0 \rightarrow$ minima

(*) $f''(x) < 0 \rightarrow$ maxima

ML $\Rightarrow$ minima



Gradient Descent $\Rightarrow$ optimization

(*) maxima | minima

(1) initialize variable x Randomly

(2) Repeat

{

$x = x - \eta \frac{\partial L}{\partial x}$

}

Loss

ascend

descent

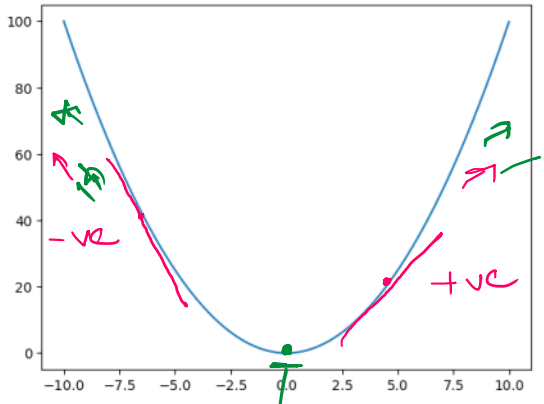
∇L

(*) Gradient Direction

Gradient Direction ?

↳ steepest

$\left[\frac{\partial L}{\partial x}\right]$



→ outwards

→ going Right

value is ↑

minima

outward

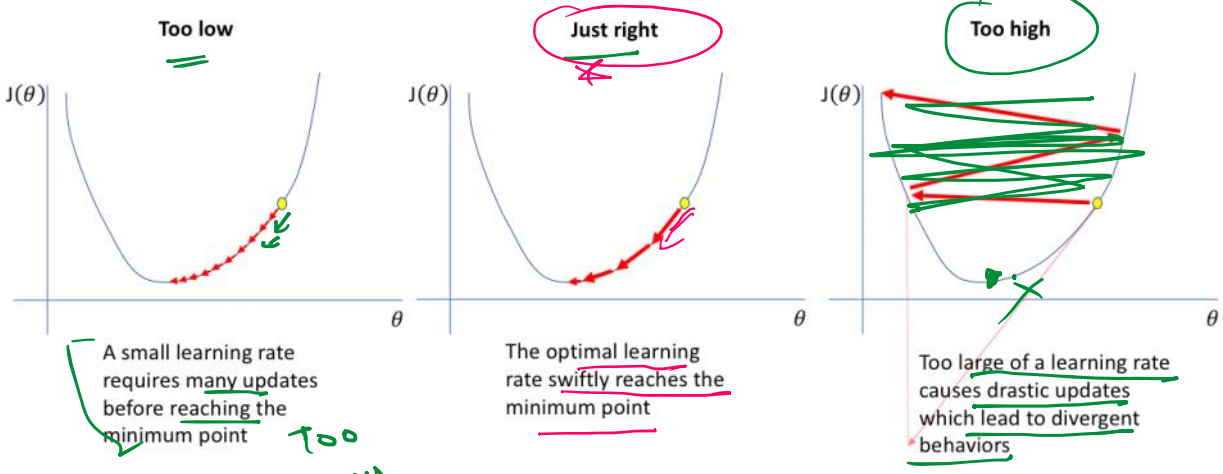
⇒ inwards

− gradient

Gradient Descent

Learning Rate ⇒ η

$\eta = 0.1$



Too many updates

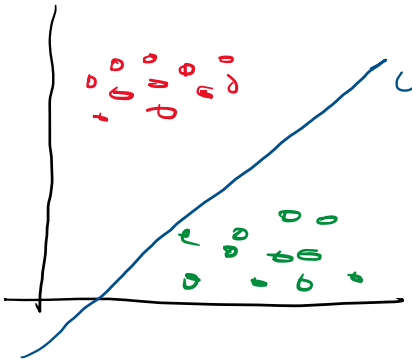
many updates

Training time

$\leftarrow$ [0.1 – 1] $\rightarrow$

ML $\Rightarrow$ [Hyperparam tuning]

∞ Possible lines



$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$ w_0

↓

gain funⁿ

or

loss funⁿ

↓

w & w_0

↳ minimize the loss funⁿ

$$w_1^*, w_0^* = \underset{w_1, w_0}{\text{argmin}} \text{ loss funⁿ}$$

optimization problem

↓

gradient descent

⇒ Compute gradients / VECTOR CALCULUS

(1) Plane (4)

$$f(x_1, x_2, x_3) = \vec{a}^T \cdot \vec{x}$$

↳ dot product

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$f(x_1, x_2, x_3) = a_1 x_1 + a_2 x_2 + a_3 x_3$$

$\nabla_x f(\vec{x}) \Rightarrow$ $a_1 x_1 + a_2 x_2 + a_3 x_3$

↳ grad. of func^ wrt x

multiple → partial

$$\nabla_x f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \vec{a}$$

$\frac{\partial f}{\partial x}$ → (∇x) = (2)

$\frac{\partial f}{\partial x} = (2)$

$=$

$$\boxed{\nabla_{\vec{x}} f(\vec{x}) = \vec{a}}$$

(2) $f(\vec{x}) = x^T \cdot x$

$x_1^2 + x_2^2 + x_3^2 + \dots x_d^2$

$$f(x_1, x_2, x_3 \dots x_d) = \sum_{i=1}^{d} x_i \cdot x_i$$

$$= \sum_{i=1}^{d} x_i^2$$

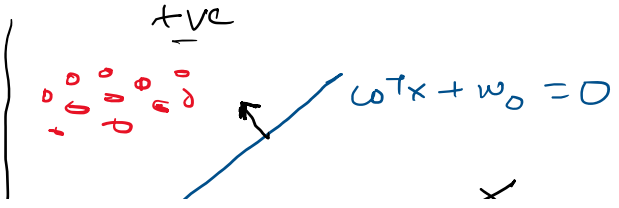
$$\nabla_x f(\vec{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_d \end{bmatrix} = 2\vec{x}$$

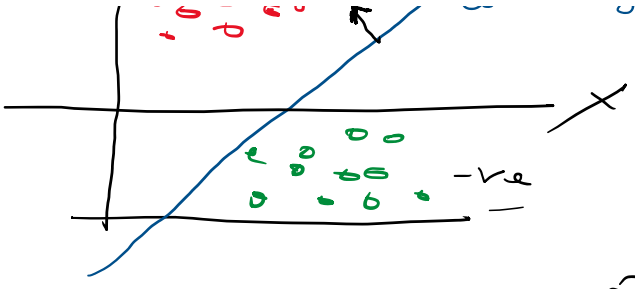
$$\boxed{\nabla_x f(\vec{x}) = 2\vec{x}}$$

goal → optimization problem

↓

Line





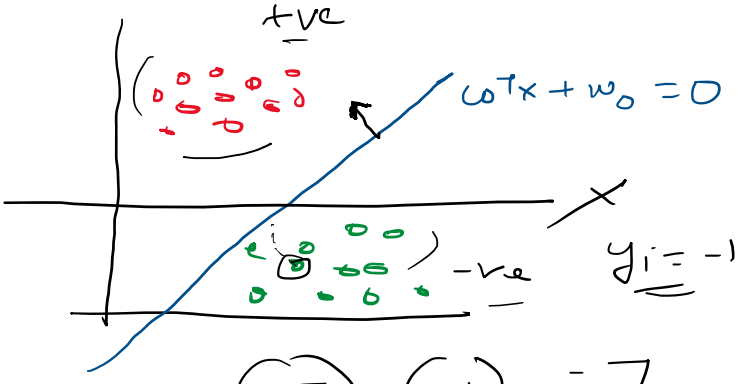
good or Bad

(*) quantity ⇒ (*) avg. dist ⇒ max

(a) $$gain(\vec{w}, w_0) = \frac{1}{n}\sum_{i=1}^{n} \frac{(w^T x_i + w_0)}{\|w\|} \cdot y_i$$

signed distance

(b) +ve



$(-7) \times (-1) = 7$

(*) → gain is maximised

$w, w_0 \rightarrow$ gain ⇒ quantity

$$w^*, w_0^* = \arg\max_{w, w_0} \text{gain}(w, w_0)$$

$$\text{loss}(w, w_0) = -\frac{1}{n} \sum_{i=1}^{n} \left(\frac{w^T x^i + w_0}{\|w\|} \right) \cdot y^i$$

$$w^*, w_0^* = \arg\min_{w, w_0} -\frac{1}{n} \sum_{i=1}^{n} \left(\frac{w^T x_i + w_0}{\|w\|} \right) \cdot y_i$$

optimization fun^n

Best values

multiple vars

Gradient Descent

$$x^* = \arg\min_{x} (x-5)^2$$

(1)

(2)

Line

(2)

New Section 9 Page 8

$$w^*, w_0^* = \arg\min_{w, w_0} \frac{1}{n} \sum_{i=1}^{n} (w^T x^i + w_0) \cdot y_i$$

$\|w\|$ → $\sqrt{w_1^2 + w_2^2}$

$w \to 100$ → $\sqrt{w_1^2 + w_2^2 + \cdots + w_{100}^2}$

Next step → finding Best Classifier

→ grad. Descent → constant

(X) → data

(1) Randomly init w & w_0 ⇒ → Line

(2) Repeat {

$w = \begin{Bmatrix} w_1 \\ w_2 \end{Bmatrix}$

$$\vec{w} = w - \eta \frac{\partial L}{\partial \vec{w}}$$

w_2

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

}

→ Suppose denominator does not exist

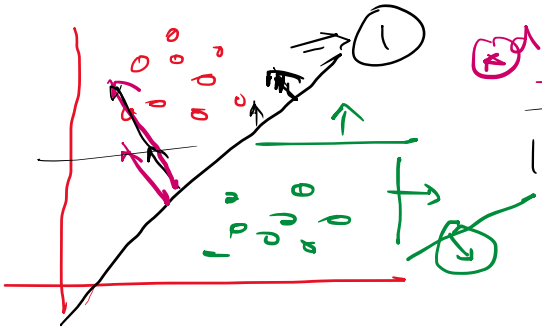
$\|w\|$ → distance → unconstrained

$\sqrt{w_1^2 + w_2^2 + \cdots + w_{100}}$

$$w^*, w_0^* = \underset{w, w_0}{\operatorname{argmin}} -\frac{1}{n}\sum_{i=1}^{n} (w^T x_i + w_0) \cdot y_i$$

$\|w\|$ ✗

$\|w\| = 1$ → magnitude ✗



direction ✓

$\|w\| = 1$ $\|w\|$

Length = 1

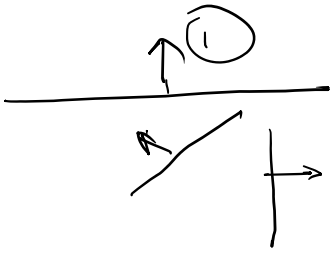
$\vec{w}$ = giving line & dir"

magnitude ✗

↓

dir"

if $\|w\| = 1$



Better Reason

=

optimization

denominator ⇒ ✗

time

gradient → Becomes easier

ML ⇒

$\|w\| = 1$ ⇒

Loss fun^{n} $\Rightarrow$ $\frac{-1}{n}\sum_{i=1}^{n}\frac{(w^T x_i + w_0)\cdot y_i}{\|w\|}$

such that $\|w\| = 1$

distance $\rightarrow$ $\|w\|$ ✗

interest $\rightarrow$ direction

$\Rightarrow$ CONSTRAINT optimization

Constraint $\rightarrow$ $\|w\| = 1$ ($\|w\|$)

opt

updated eq^{n}

$$L(w, w_0) = \frac{-1}{n}\sum_{i=1}^{n}(w^T x^i + w_0)\cdot y_i$$

such that $\|w\| = 1$ (*)

(*) $\vec{w}$ $\rightarrow$ direction

To solve constraint optimization problem $\Rightarrow$

⇒ LAgRange multiplier

CONSTRAINT optimization PROBLEM ⇒

=

q → optimise → $y = x^2 - 3x - 3$

$1^2 - 3 - 3$

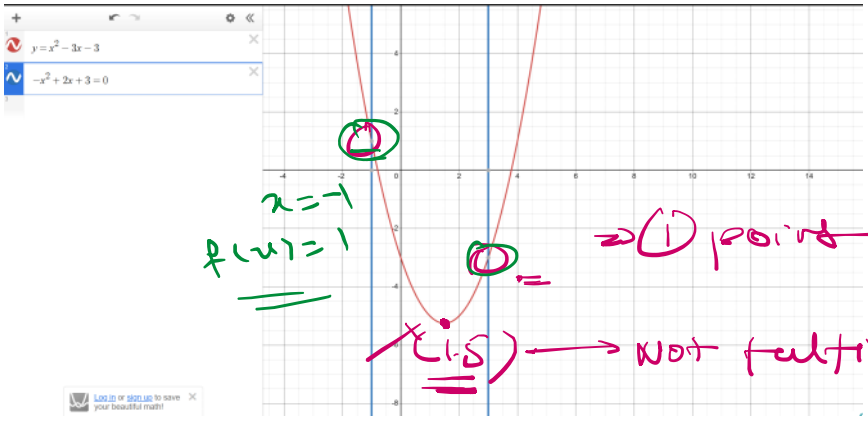
→ minima

↳ such that

$-x^2 + 2x + 3 = 0$ (*)

CONSTRAINT

$-(1.5)^2 + 2(1.5) + 3$



x=-1

f(x)=1

⇒ (1) point

x(1.5) → NOT fulfilling constraint

⇒ minima ?

x=3

f(x) = 3

⇒ constraint fulfilled

x=3 → minima

Break → 10:27

$x = -1 \Rightarrow (+1)^2 - 3(-1) - 3$

$\Rightarrow 1 + 3 - 3$

$= 1$

$-x^2 + 2x + 3 = 0$

$\Rightarrow -x^2 + 3x - x + 3 = 0$

$\Rightarrow -x(x+3) - 1(x-3) = 0$

$(-x-1)(x-3) = 0$

$x = -1, 3$

1 min

optimization

Boost → a lot of time

Paytm → Scan ⇒

Fraud detection

< 500 ms ✓

Precision Recall

Precision Recall

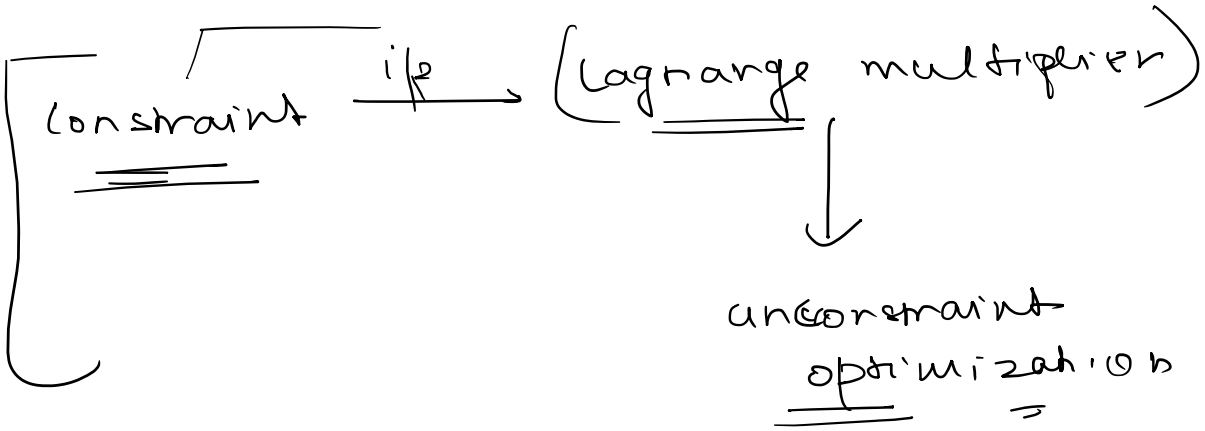
Maths → ✓

✗ → implements

Packages → Scikit

Param.

(✗) constraint opt. problem →



Unconstraint opt. problem →

we have seen so far

↓

→ Unconstrained

↓

easy → QP

opt. proble

Lagrange multiplier →

argmin f(x) such that g(x) = 0
x

↓ constraint (1)

x, λ ⇒ argmin f(x) + λ · g(x)
x, λ

x, λ → variable

λ → Lagrange multiplier

↳ (n+1) (2)

100 = # constraints ⇒

argmin f(x) such that
x

$g_1(x) = 0$

$g_2(x) = 0$

$g_3(x) = 0$

⋮

$g_{100}(x) = 0$

100 constraints



argmin $f(x) + \lambda_1 g_1(x) + \lambda_2 g_2(x) + \dots + \lambda_{100} g_{100}(x)$

$x, \lambda_1, \lambda_2 \dots \lambda_{100}$

↳ intermediate vars

↳ help in BLO

$(n+1) \Rightarrow$ to optimise



Constraint opt. → Lagrange → unconstraint



$q \rightarrow \underset{x}{\text{argmin}}\ x^2 - 3x - 3 = 0$

such that

$-x^2 + 2x + 3 = 0$

↓

unconstraint

unconstraint

$$= \arg\min_{x,\lambda} (x^2 - 3x - 3) + \lambda(-x^2 + 2x + 3)$$

$\rightarrow x^*, \lambda^*$

↓

(1) gradient Descent

GD says →

(1) x & λ → Randomly

(2) update → $x = x - \eta \times \frac{\partial L}{\partial x}$

$\lambda = \lambda - \eta \times \frac{\partial L}{\partial \lambda}$

(4)

$x = x - \eta \cdot \frac{\partial L}{\partial x}$ ⇒

$$= \arg\min_{x,\lambda} (x^2 - 3x - 3) + \lambda(-x^2 + 2x + 3)$$

$\rightarrow x^*, \lambda^*$

$$\frac{\partial L}{\partial x} \Rightarrow [2x - 3 - 0 - 2\lambda x + 2\lambda + 0]$$

$$= 2x - 3 - 2\lambda x + 2\lambda \quad (1)$$

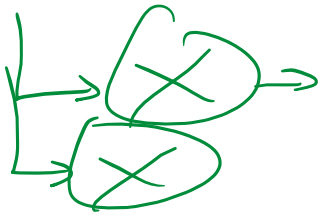
(1) $\frac{\partial L}{\partial \lambda} \Rightarrow 0 + 0 + 0 - x^2 + 2x + 3$

Repeat this 100 times

Result

minima

x new, λ new



$$\frac{\partial L}{\partial x} = 0 \Rightarrow 2x - 2\lambda x + 2\lambda - 3 = 0 \quad (1)$$

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow -x^2 + 2x + 3 = 0 \quad (2)$$

$x, \lambda \rightarrow$ +ve

$$-x^2 + 2x + 3 = 0$$

$x = -1, x = 3$

only possible values of x

Understand

Example

(*) optimisation problem

updated eq

$$L(w, w_0) = -\frac{1}{n} \sum_{i=1}^{n} (w^T x^i + w_0) \cdot y^i$$

such that $\|w\| = 1$

(*) $\vec{w}$ → direction

Constraint opt. problem

$$\sqrt{w_1^2 + w_2^2} = 1$$

$$\sqrt{w^T w} = 1 \quad (*)$$

$$\hookrightarrow \sqrt{w^T w} - 1 = 0$$

$$w^*, w_0^*, \lambda^* =$$

$$\underset{\vec{w}, w_0, \lambda}{\text{argmin}} = \frac{-1}{n} \sum_{i=1}^{n} (w^T x^i + w_0) \cdot y_i + \lambda \cdot \left(\sqrt{w^T w} - 1 \right)$$

$\vec{w}$ w_0 λ

$\vec{w} \rightarrow w_1, w_2$

min - value

↓

apply GD

→ Randomly w, w_0, λ

→ Repeat {

$$w = w - \eta \nabla_w L \Rightarrow$$

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}$$

$$\lambda = \lambda - \eta \frac{\partial L}{\partial \lambda}$$

$$\lambda = \lambda - \eta \frac{\partial L}{\partial \lambda}$$

Loss

$$= \frac{-1}{n} \sum_{i=1}^{n} (w^T x^i + w_0) \cdot y^i + \lambda (w^T w - 1)$$

$w_0 \perp$ 0

$$\frac{\partial L}{\partial w_0} = \frac{-1}{n} \sum \left[\frac{\partial}{\partial w_0} (w^T x_i y_i) + \frac{\partial}{\partial w_0} (w_0 y_i) \right] = 0$$

$$= -\frac{1}{n} \sum_{i=1}^{n} y_i$$

Tricky

$\frac{\partial L}{\partial w_1}$ $\frac{\partial L}{\partial \lambda}$

⇒ Solve on Monday

$$f(x_1, x_2) = x_2 e^{x_1} - 4 x_2^2$$

$x_2 \rightarrow$ func$^n \rightarrow x_1$

$$\frac{d}{dx_1} (x_2 \cdot e^{x_1}) = x_2 e^{x_1} + x_2 \cdot e^{x_1}$$

$$\frac{d}{dx_1}(x_2 \cdot e^{x_1}) = x_2 e^{x_1} + x_2 \cdot e$$

$$\frac{d}{dx}(-4x_2^2) = -8x_2$$

(2) $f = x_2 \cdot e^{x_1} - 4x_2^2$

(1) (2)

$$\frac{d}{dx_1}(u \cdot v) = u \cdot v' + u \cdot v$$

$u = x_2$, $v = e^{x_1}$

$$f = x_2 \cdot e^{x_1} - 4x_2^2$$

$$f(x_1, x_2) = x_2 e^{x_1} - 4x_2^2$$

(2)

$\frac{df}{dx_1} \Rightarrow x_1$

$x_1' \Rightarrow \frac{dx_1}{dx_2}$

$\frac{df}{dx_1}$

x_2' ⇒ $\frac{dx_2}{dx_1}$

x_2 const → $e^{x_1} - 8x_2$

$$\frac{df}{dx_1} = \frac{df}{dx_1} + \frac{df}{dx_2} \cdot \frac{dx_2}{dx_1}$$

⇒ x^2 as const

⇒ $x_2 \cdot e^{x_1}$

⇒ confusing

Checked → Next class